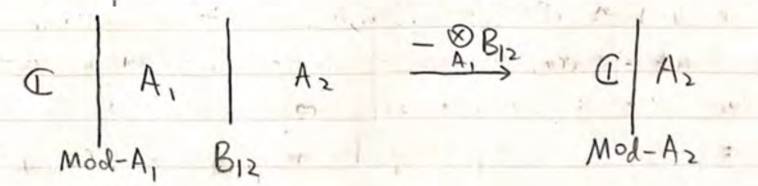
Ohmori

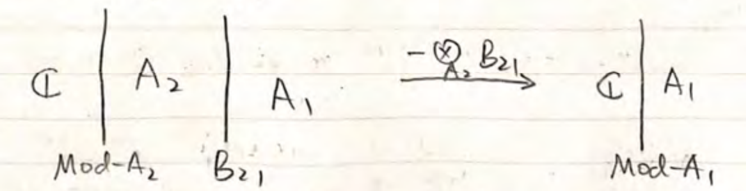
6/17

①

Last lecture, we knew that two phases are equivalent if they are Morita equivalence. So we need to study Morita class more carefully $A_1 \overset{\text{Morita}}{\simeq} A_2$ iff $\text{Mod-}A_1 \simeq \text{Mod-}A_2$. For general category, Mod-A is too difficult to compute. But, for SpFrob $\text{Mod-}A \simeq 1\text{-Hom}_{\text{SpFrob}}(\mathbb{C}, A)$ which is computable. To see $A_1 \overset{\text{Morita}}{\simeq} A_2 \Rightarrow \text{Mod-}A_1 \simeq \text{Mod-}A_2$:

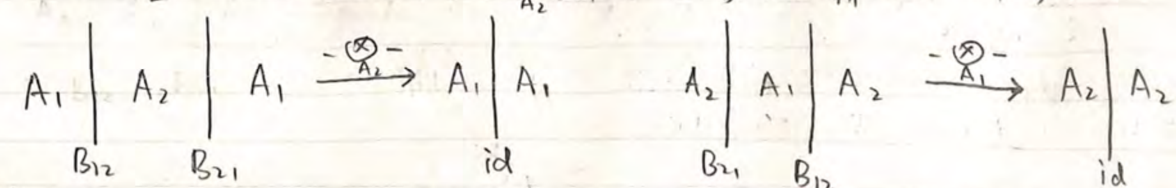


$-\underset{A_1}{\otimes} B_{12} : \text{Mod-}A_1 \to \text{Mod-}A_2$



$-\underset{A_2}{\otimes} B_{21} : \text{Mod-}A_2 \to \text{Mod-}A_1$

Morita equivalence means $B_{12} \otimes_{A_2} B_{21} \simeq A_2$, $B_{21} \otimes_{A_1} B_{12} \simeq A_1$, i.e.



That is we have invertible functor $-\otimes_{A_1} B_{12}$ and $-\otimes_{A_2} B_{21}$ between Mod-A_2 and Mod-A_1, which means they are equivalent.

How Mod-A looks like? We knew thate $\text{SpFrob} \ni A \simeq \bigoplus_{t=1}^{\tilde{N}} \text{Mat}_{n_t \times n_t}$
$\Rightarrow \text{Mod-}A \ni E_t \simeq \mathbb{C}^{n_t}$. Any Module M decomposes as $M \simeq \bigoplus_{t=1}^{\tilde{N}} (E_t)^{\oplus m_t}$
Here $m_t \in \mathbb{Z}_{\geq 0}$ and

$$\text{Hom}_{\text{Mod-}A}(E_{t_1}, E_{t_2}) = \begin{cases} \mathbb{C}, & t_1 = t_2 \\ 0, & t_1 \neq t_2 \end{cases}$$

Which means Mod-A is a finite semi-simple linear Category.
semi-simple means $\exists$ a basis in Ob C $\{S_1, \cdots\}$ s.t. any object can be spanned by them:

$$\text{ObC} \ni A = \bigoplus_{t=1}^{N} \mathbb{C}^{m_t} \underset{\text{Vect}_{\mathbb{C}}}{\otimes} S_t$$

Finite means the length of this basis is finite. We can chose a orthogonal basis which means

$$\text{Hom}_C(S_{t_1}, S_{t_2}) = \begin{cases} \mathbb{C}, & t_1 = t_2 \\ 0, & t_1 \neq t_2 \end{cases}$$

②

Then, $\text{Hom}_C(A, \tilde{A}) = \bigoplus_{t=1}^{N} \bigoplus_{s=1}^{N} (\mathbb{C}^{m_t} \otimes \mathbb{C}^{n_s}) \otimes \text{Hom}_C(S_t, S_s) = \bigoplus_{t=1}^{N} \mathbb{C}^{m_t n_s}$

In the above disscussion we also implied linear 1-cat structure.
which means A is $\text{Vect}_{\mathbb{C}}$-enriched:

$\underset{\text{Vect}_{\mathbb{C}}}{\otimes} : \text{Vect}_{\mathbb{C}} \times C \to C$ $\qquad$ $\text{Hom}_C(A_1, A_2)$ is a vector space
$\qquad \mathbb{C} \times A \mapsto A$ $\qquad$ and has the following structure

$$\text{Hom}_{\text{Vect}_{\mathbb{C}}}(\mathbb{C}, \mathbb{C}) \times \text{Hom}_C(A_1, A_2) \simeq \mathbb{C} \times \text{Hom}_C(A_1, A_2) \xrightarrow{\otimes_{\text{Vect}_{\mathbb{C}}}} \text{Hom}_C(A_1, A_2)$$

Get rid of these roughly mathmatical stuff, back to oure main road.
it's not hard to see finite-sim-simple linear category C is

$$C \simeq \text{Vect}_{\mathbb{C}}^{\boxplus N}, \quad N \text{ is the \# of Simple object (\# of } S_i)$$

This can be though as an analog of $V \simeq \mathbb{C}^N$ in $\text{Vect}_{\mathbb{C}}$ category.
For example, consider a finite Group G, $\mathbb{C}[G] := \text{Span}_{\mathbb{C}}\{g_1, \cdots g_{|G|}\}$

$$\text{Mod-}\mathbb{C}[G] \simeq \text{Rep}(G) \simeq \text{Vect}^{\boxplus(\#\text{ of irreps})}$$

Here, we ignore the $\otimes$-structure on Rep(G). Therefore:

$$\mathbb{C}[G_1] \overset{\text{Morita}}{\simeq} \mathbb{C}[G_2] \text{ if \# of irreps matches}$$

Actually, $\mathbb{C}[G] \simeq \bigoplus_{\rho \in \text{irrep}} \text{End}_{\mathbb{C}}(\rho)$

Now, we obtained a 2-functor

$$\begin{aligned} \text{Mod} : \text{SpFrob} &\longrightarrow \text{fin. s.s. linear categories} \\ A &\longmapsto \text{Mod-}A \quad (\#\text{ of ground states}) \\ B_{12} &\longmapsto -\underset{A_1}{\otimes} B_{12} \\ f &\longmapsto \varphi^f : -\underset{A_1}{\otimes} B_{12} \Rightarrow -\underset{A_1}{\otimes} \tilde{B}_{12} \quad \varphi^f_x := \text{id}_x \underset{A}{\otimes} f \end{aligned}$$

natural transformation

More over, Mod is a 2-equiv. of 2-Cat.

- Tannaka-Klein reconstruction

How to get A from a fin. s.s. linear 1-Cat, C s.t. $\text{Mod}(A) \simeq C$

Step 1: pick $G \in \text{Ob}\, C$ "big enough", i.e. $\text{Hom}_C(G, X) \neq 0$ if $X \neq 0$
since C is a fin. s.s. linear 1-cat

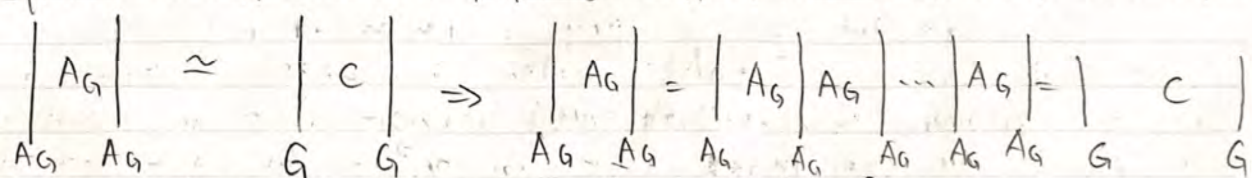
$$G = \bigoplus_t^{\tilde{N}} \mathbb{C}^{g_t} \otimes S_t, \quad g_t \geq 1$$

Step 2: $A_G := \text{Hom}_C(G, G) \simeq \bigoplus_{t=1}^{\tilde{N}} \text{Mat}_{g_t \times g_t}$

Then $\text{Mod}(A_G) \simeq C$ by functor $F_G(x) := \text{Hom}_C(G, X)$

③

Step 3: $B_{12} := \mathrm{Hom}_C(G_1, G_2)$ is a (A_{G_1}, A_{G_2})-bimodule and it is invertible which means $A_{G_1} \overset{\text{Morita}}{\simeq} A_{G_2}$, so A is independent on G.
So G parametrizes algebra (lattice model) in the Morita class (phase). So different choice of G is just different lattice realization for the same phase.



i.e. $A_G \simeq A_G \otimes_{A_G} A_G \cdots \otimes_{A_G} A_G \Rightarrow \mathcal{H}_{total} = (A_G)^{\otimes L}$

$$\mathcal{H}_{total} = \underset{G\quad G}{\overset{C}{\vdash\!\!\dashv}}\ \underset{G\quad G}{\overset{C}{\vdash\!\!\dashv}} \cdots \underset{G\quad G}{\overset{C}{\vdash\!\!\dashv}}$$

The analog in QM is

$$\langle\phi|\psi\rangle = \langle\phi|\mathbb{1}\cdot\mathbb{1}\cdots\mathbb{1}|\psi\rangle = \sum_{t_1\cdots t_N}\langle\phi|e_{t_1}\rangle\langle e_{t_1}|e_{t_2}\rangle\cdots\langle e_{t_N}|\psi\rangle$$

Here $\underset{G\,G}{\dashv\vdash}$ can be seen as $\sum |G\rangle\langle G|$ "higher complete set". So we just use G to discretise the spatial direction, then we obtain a lattice model of one continuous phase, different G gives you different discretization i.e. different lattice model but since they are Morita equivalent to each other, so they describe a same phase. And the Hamiltonian can be constructed by the same method as before, from MPO Projector $P_{i,i+1}$

§ Karoubi Completion

Now we turn to 2+1d theory. Following the notation in 1905.09566 [GJF]

$$\left.\begin{array}{lll} & \mathrm{Vect}_{\mathbb{C}} & 0+1\,d \\ \mathrm{Sp\,Frob} \rightsquigarrow & 2\text{-}\mathrm{Vect}_{\mathbb{C}} & 1+1\,d \end{array}\right]\ \text{we have learned}$$

$\vdots$

$n\text{-}\mathrm{Vect}_{\mathbb{C}}$ $\quad (n-1)+1\,d$

GJF proposal: $n\text{-}\mathrm{Vect}_{\mathbb{C}} := \mathrm{Kar}(B\,(n-1)\text{-}\mathrm{Vect})$

Kar means karoubi completion. B denotes to delooping. For a monoidal 1-cat $(\mathcal{V}, \otimes)$ $B\mathcal{V}$ is a 2-Cat with one obj $\{*\}$ and

$1\text{-}\mathrm{Hom}_{B\mathcal{V}}(*,*) := \mathcal{V}$

The composition is given by monoidal structure of $\mathcal{V}$, $\otimes: \mathcal{V}\times\mathcal{V} \to \mathcal{V}$

④

Similarly, for a n-cat $\mathcal{V}$, $B\mathcal{V}$ gives a $(n+1)$-cat. If $\mathcal{V}$ is a symmetric monoidal category, then $B\mathcal{V}$ is also a symmetric monoidal category, as well as $B^k\mathcal{V}$

For simplicity, first consider 1-Cat, a morphism $p \in \mathrm{Hom}_C(X,X)$ is idempotent iff $p^2 = p$. And an idempotent p is split iff $\exists Y \in \mathrm{obj}(C)$, $r: X \to Y$, $s: Y \to X$ s.t. $s\circ r = p$, $r\circ p = \mathrm{id}_Y$. C is karoubi complete iff every idempotent is split. For any 1-Cat C we can make it karoubi complete by the following construction

$$\mathrm{Obj}(\mathrm{Kar}(C)) = \{(X, p) \mid X \in \mathrm{Obj}(C),\ p: \text{idempotent on } X\}$$

$$\mathrm{Hom}_{\mathrm{Kar}(C)}((X, P_X), (Y, P_Y)) = \{f: X \to Y \mid f\circ p_X = f = P_Y \circ f\}$$

Hence we have a natural embedding

$$C \hookrightarrow \mathrm{Kar}(C) : X \mapsto (X, \mathrm{id}_X)$$

Since $\mathrm{id}_X^2 = \mathrm{id}_X$. It's not hard to see if C is already karoubi complete then $\mathrm{Kar}(C) \simeq C$, i.e. any terms we add in C is already included in C.

e.g. $F := \bigcup_{n=1}^{\infty} \mathrm{Mat}_{n\times n} = \{(\infty\times\infty)$ matrices with finitely many nonzero elements$\}$

Notice: $\mathbb{1} \notin F$, so F is a non-unital alg. (it's more like "0-vect")

BF is also a non-unital 1-Cat and

$$\mathrm{Kar}(BF) \xrightarrow{\sim} \mathrm{Vect}_{\mathbb{C}}^{fin}, \quad (*, P_{k\times k}) \mapsto \mathbb{C}^k$$

Here $P_{k\times k} \in F$ means

$$P_{k\times k} = \begin{bmatrix} \underbrace{\begin{matrix} 1 & & \\ & \ddots & \\ & & 1 \end{matrix}}_{k}\Big\}k & & & \\ & 0 & & \\ & & 0 & \\ & & & \ddots \\ & & & & 0 \end{bmatrix}_{\infty\times\infty} \in F$$

This story continues on higher categories. Considering a 2-cat, C, $A \in 1\text{-}\mathrm{Hom}_C(X,X)$ $\square$ denotes the composition in 1-Hom. Naively, we may require $A \square A = A$ to be an idempotent, but for $C = B\mathrm{Vect}$ which A satisfies $A\square A = A$ is only $\mathbb{C}$! So it's so boring. GJF proposed for 1-cat we require $p^2 = p$, but for 2-cat we only require $A^2 \underset{\Delta}{\overset{\mu}{\rightleftarrows}} A$, $\mu\circ\Delta = \mathrm{id}_A$ and other consistency called Condensation Alg. Actually A can be considered as a Sp Frob Alg. On X. Then we can define Karoubi completion as before:

$\mathrm{Obj}(\mathrm{Kar}(C)) \ni \{(X, A) \mid X \in \mathrm{obj}(C),\ A:$ Sp. Frob. Alg on $X\}$
And some other conditions on Hom set. For $C = \mathrm{Vect}$, we have

$$2\text{-Vect} = \mathrm{Kar}(B\mathrm{Vect}) = \mathrm{SpFrob}.$$

With these prerequisite, we can talk about 2+1d gapped phase, i.e

$$3\text{-Vect} = \mathrm{Kar}(B2\text{-Vect}) = \mathrm{Kar}(\mathrm{SpFrob})$$

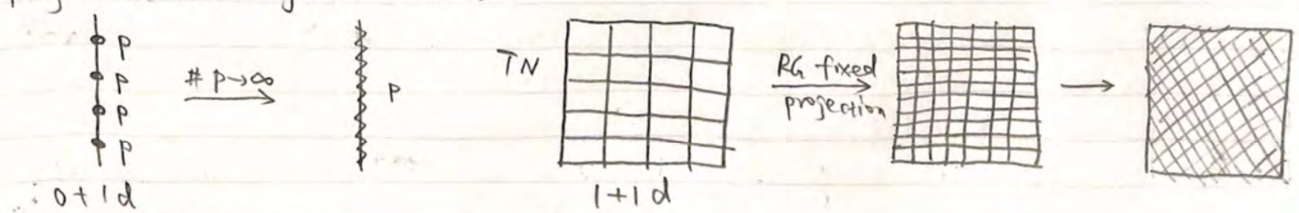
The rigorous definition of Karoubi completion of 3-Cat is too technical to explain. However, we can guess it from some experience in Kar(1-Cat) and Kar(2-Cat). First, the shape of $\mathrm{Obj}(\mathrm{Kar}(B2\text{-Vect}))$ is $(X, \mathbb{A})$. From the experience on Kar(2-Cat) $\mathbb{A}$ is something like Sp Frob Alg. But μ, Δ should be replaced to Higher version:

$$\mathbb{A}^2 \underset{\Delta^{(2)}}{\overset{\mu^{(2)}}{\rightleftarrows}} \mathbb{A}, \quad \mu^{(2)} \circ \Delta^{(2)} \underset{s}{\overset{r}{\rightleftarrows}} id_{\mathbb{A}}, \quad r \circ s = id_{id_{\mathbb{A}}}$$

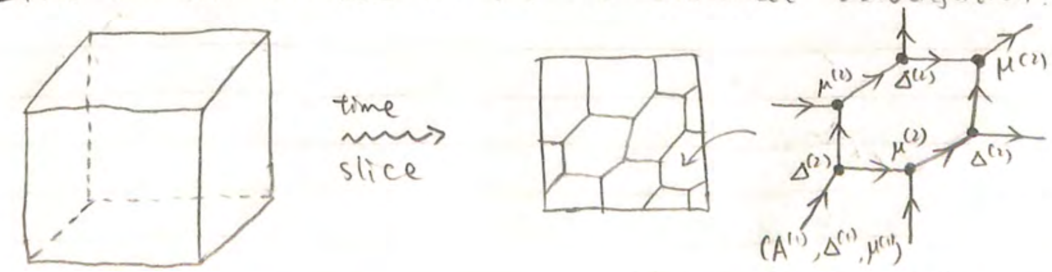
$\mu^{(2)}$ and $\Delta^{(2)}$ are in 2-Hom of 3-Cat i.e. they are bi-modules, and $\mathbb{A}$ itself is a Sp Frob Alg $(A^{(1)}, \mu^{(1)}, \Delta^{(1)})$, they have the structure as an idempotent in 2-Cat i.e. $(A^{(1)})^2 \underset{\Delta^{(1)}}{\overset{\mu^{(1)}}{\rightleftarrows}} A^{(1)}$, $\mu^{(1)} \circ \Delta^{(1)} = id_{A^{(1)}}$

And, $\mu^{(2)} = (A^{(1)} \otimes_{\mathbb{C}} A^{(1)}, A^{(1)})$ − bimodule
$\Delta^{(2)} = (A^{(1)}, A^{(1)} \otimes_{\mathbb{C}} A^{(1)})$ − bimodule

physical view of Karoubi completion is condensation i.e. adding projection onto ground states.



The time slice in 1+1d gives the Hilbert space we have mentioned $\mathcal{H}^{tot} = A^{\otimes L}$. and the image of projection is ground states Hilbert space
In 2+1d we can also use 3d TN to construct condensation.



2+1d

Y-junction of $A^{(1)}$'s

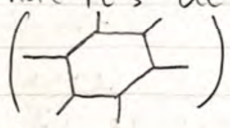
As before the time slice gives us the Hilbert space

$$\mathcal{H}^{tot} = \bigotimes_{\mathbb{C}} \Delta^{(2)} \otimes_{\mathbb{C}} \bigotimes_{\mathbb{C}} \mu^{(2)}$$

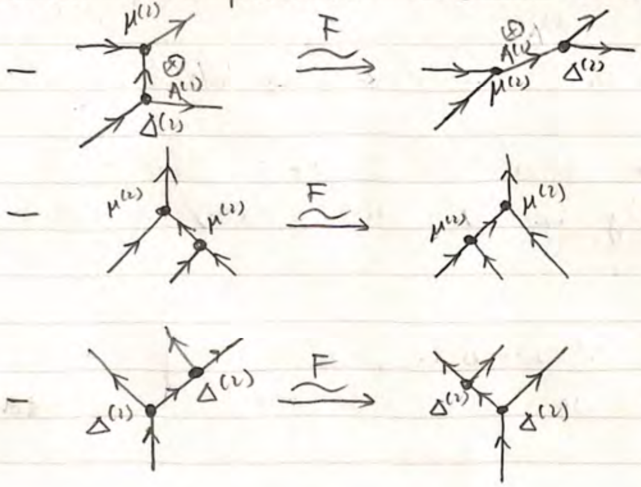
We also define tensor product of $A^{(1)}$,

$$\mathcal{H}^{(1)} := \bigotimes_{A^{(1)}} \Delta^{(2)} \otimes_{A^{(1)}} \mu^{(2)}$$

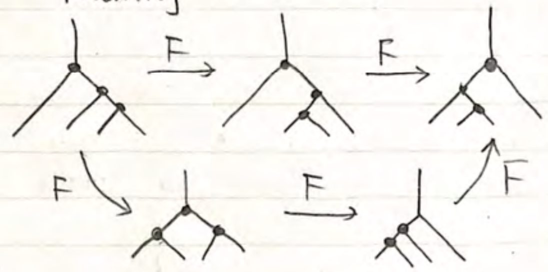
And there is a bijection between $\mathcal{H}^{tot}$ and $\mathcal{H}^{(1)}$. One important feature of $\mathcal{H}^{(1)}$ is that it's detail independ (condensable)

$\mathcal{H}^{(1)}\Big($  $\Big) \simeq \mathcal{H}^{(1)}\Big($TN with the same # of faces$\Big)$

Moves that preserve the number of faces are generated by

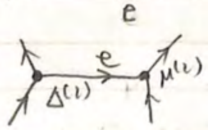


These properties of "F-move" are a higher version of Frob property. And it's easy to see F-symbol should satisfy pentagon identity

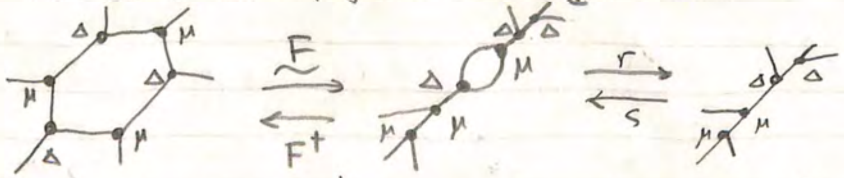


The Hamiltonian on $\mathcal{H}^{tot}$ is (like toric code model)

$$H = -\sum_e P_e - \sum_p P_p, \quad e: \text{edge} \quad p: \text{plagutte (face)}$$

P_e :  $\quad \Delta^{(2)} \otimes_{\mathbb{C}} \mu^{(2)} \underset{p}{\overset{p^\dagger}{\leftrightarrows}} \Delta^{(2)} \otimes_{A^{(1)}} \mu^{(2)}$

then $P_e := p \circ p^\dagger$ as a projection on $\Delta^{(2)} \otimes_{\mathbb{C}} \mu^{(2)}$ to $\mathcal{H}^{(1)}$

P_p : 

then $P_p := F \circ r \circ s \circ F^\dagger$ as a projection from $\Delta^{(1)} \otimes_{\mathbb{C}} \mu^{(2)}$ to $\mathcal{H}^{(1)}$

example :
let's set $A^{(1)} = \mathbb{C}^{\oplus N} = \mathrm{span}\{e_1, \cdots e_N\}$ $\quad e_i \cdot e_j = \delta_{ij}$

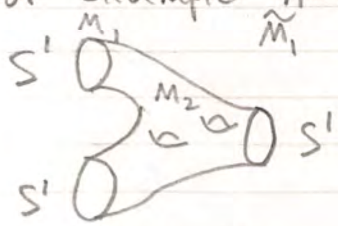
(8)

§ Cobordism hypothesis.

Non-extended TQFT [Atiyah]

$$Z = \mathrm{Bord}_{(n,n-1)} \longrightarrow \mathrm{Vect}_{\mathbb{C}}$$

For example $n=2$



$\xrightarrow{Z}$ $Z(M_1) \xrightarrow[\text{linear map}]{Z(M_2)} Z(\tilde{M}_1)$

$\uparrow$ Vect space.

$S^1 \sqcup S^1 \sqcup S^1$

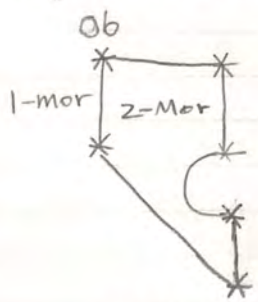
There are some axioms, the most important one is "gluing" inspired by path integral.

Similarly, we can generalize this to Fully extended TQFT

$$Z : \mathrm{Bord}_{(n,n-1,\cdots,0)} \longrightarrow \mathcal{C}$$

left hand side is a (∞,n)-cat, obj: 0-dim mfd {pt's}, 1-mor: 1-dim bord, 2-mor: bord of bord (with corners) ...

right hand side is a symmetric monoidal (∞,n)-cat



Cobordism hypothesis:

Z is determined by $Z(*)$

But it doesn't constrain on $\mathcal{C}$, physicist want to work with concrete category.

[GJF] chose $\mathcal{C} = n$-Vect

$Z(M_n) \in \mathbb{C}$

$Z(M_{n-1}) \in \mathrm{Vect}$

$Z(M_{n-2}) \in 2\text{-Vect}$

...

So to reconstruct the whole Z, we only need to know $Z(*) \in (n-1)$-Vect

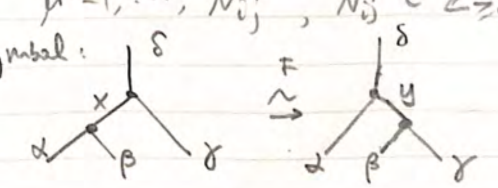
In this lecture, for $n \leq 3$, we explicitly construct it by using Tensor Network

(7)

$\mu^{(2)}$: $(A^{(2)} \otimes A^{(2)}, A^{(2)})$ – bimodule $= \bigoplus_{i,j,k} \mathbb{C}^{N_{ij}^k} \ni (m_{ij}^k)_\mu$

$\mu = 1, \cdots, N_{ij}^k$, $N_{ij}^k \in \mathbb{Z}_{\geq 0}$, and $(e_i \otimes e_j) \cdot (m^\gamma_{\alpha\beta})_\mu \cdot e_k = \delta_{i\alpha}\delta_{j\beta}\delta_{k\gamma}(m^\gamma_{\alpha\beta})_\mu$

F-symbol:

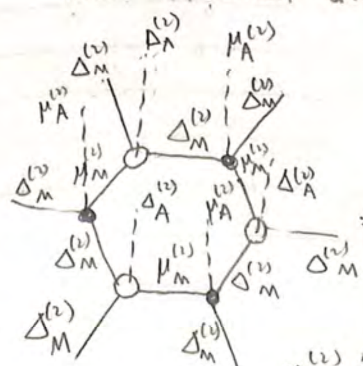


$$F((m^x_{\alpha\beta})_\mu, (m^\delta_{x\gamma})_\nu) = (F^{\delta,x,y}_{\alpha,\beta,\gamma})_{\mu,\nu,\rho,\sigma} (m^\delta_{\alpha y})_\rho (m^y_{\beta\gamma})_\sigma$$

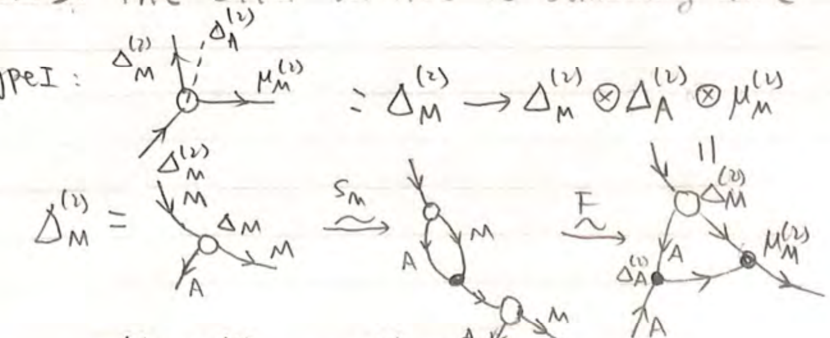
We require pentagon identity on $(F^{\delta,x,y}_{\alpha,\beta,\gamma})_{\mu,\nu,\rho,\sigma}$. From above disscussion $\mathrm{Mod}\text{-}A^{(1)} \simeq \mathrm{Vect}_{\mathbb{C}}^{\boxplus N}$ gets the structure of monoidal cat. $(\Delta^{(2)}, \mu^{(2)} \underset{(A^{(1)})^2}{\otimes} \Delta^{(2)} \rightleftarrows A^{(1)})$ ⇝ Fusion category ⇝ Levin-Wen *

More generally, for a fusion 1-Cat. And choose $G \in f$ by TK construction, we can obtain $A^{(1)} = \mathrm{Hom}_f(G,G)$ gets $(\mu^{(2)}, \Delta^{(2)}, F, \cdots)$ forming $A \in 3$-Vect. If we chose $f = \mathrm{Rep}(H)$ and G as regular rep $\mathbb{C}[H]$. Actually we will find Kitaev g-double model

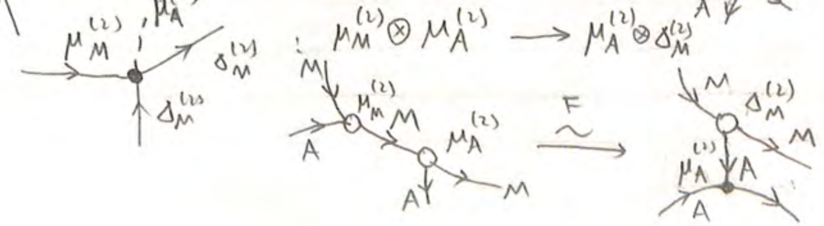
In the end, back to Tensor Networks. All of things we have done in abstruct way can be rederived from Tensor Network. Just as we did for 1+1d, we need some condition for the TN is RG-fixed, it turns out we require SpFrob structure. In 2+1d this approach also works. The 2+1d lattice is something like



type I: $\Delta_M^{(2)} \to \Delta_M^{(2)} \otimes \Delta_A^{(2)} \otimes \mu_M^{(2)}$



type II: $\mu_M^{(2)} \otimes \mu_A^{(2)} \to \mu_A^{(2)} \otimes \delta_M^{(2)}$



These give us consistent conditions similar to 1+1d case

* We only require Fusion structure i.e. F-symbol here, without assuming Braiding structure, which is necessary for cat. of anyons. It seems we can obtain the cat of anyons by taking Drinfeld center (not understood